

Curriculum Vitae

Andrew Mcveigh

Technologies

Languages

Bash, CSS, Clojure, ClojureScript, Elm, HTML, Haskell, Idris, Java, JavaScript, Python, Ruby, Rust, SPARQL, SQL

Platforms, Tools & Frameworks

AWS, Azure, Docker, EC2, ECS, Geospatial, Git, Google Cloud, Kafka, Kinesis, Linked Data, Mapbox Vector Tiles, MongoDB, PostgreSQL, RDF, RabbitMQ, React, Redis, Shapefiles, Svelte

Experience

Director / Consultant / Software Engineer: Modfacto LTD [2025 - Present]

Consulting on multiple projects for government and public sector clients, using a diverse technology stack including Python, Java, Clojure, JavaScript, React, Linked Data, RDF, SPARQL.

Developing audio applications in Rust, responsible for system design and implementation with a focus on performance, reliability, and efficient low-level processing.

Principal Engineer: Swirrl IT LTD / TPXimpact [2019 - 2024]

Joined Swirrl IT Ltd in 2019; continued in the role following acquisition by TPXimpact in 2022. Joined as a Senior Software Engineer and progressed to Principal Software Engineer.

Worked across web, geospatial, and linked-data systems using JavaScript/React, Java/Clojure backends, Clojure/Python data processing and ETL, SPARQL/RDF graph databases, PostgreSQL/PostGIS, AWS, Google Cloud, shapefiles, and Mapbox Vector Tiles.

Contributed to Swirrl's core infrastructure, improving reliability and operational maturity across cloud-hosted data platforms, including architectural decisions and deployment automation for business-critical services.

Environment Agency Programme – Lead Engineer

Lead Engineer for a long-running programme with the Environment Agency, providing technical ownership across multiple concurrent projects spanning product development and site reliability.

- Architectural oversight and engineering direction for several production systems.
- Led reliability and operational improvements for public-facing data services.
- Rewrote the Catchment Data Explorer to modernise its architecture and improve maintainability and performance.
- Took the Shoreline Management Plan platform from prototype through to full production.
- Worked closely with client stakeholders to translate complex environmental and policy requirements into robust, production-grade systems.

Senior Software Engineer: Habito [2018]

Worked primarily on backend systems and a mortgage broker portal using Haskell and PureScript, building and maintaining robust, type-safe services and user-facing functionality. Focused on improving system reliability and correctness within a functional programming paradigm.

A key achievement was diagnosing and resolving a complex, deep-seated bug in the Haskell/PostgreSQL event sourcing persistence layer, restoring data consistency and improving the stability and integrity of the platform's broker/customer data workflows.

Developer: GoMore [2017 - 2018]

GoMore is a peer-to-peer transportation startup based in Copenhagen.

Led infrastructure projects as part of the backend team, built Clojure APIs and worked on the Rails website as part of the broader product team.

Moved the product from Elastic Beanstalk to containers and migrated the main database load to Aurora, improving scalability and cost efficiency.

Developer: uSwitch Ltd. [2015 - 2017]

Platform Maintained core energy services, including the comparison engine, data APIs, processing pipelines, and data delivery. Built APIs for third-party integrations and worked on scaling/performance for 10x traffic spikes during TV promotion.

Back office Maintained backend services for day-to-day energy operations in a cross-functional team, involving commercial, operations, financial and software development. Worked on third-party data exchange, product tracking, financial reporting, and analysis.

Software Developer: Interel Group, Brussels [2008 - 2015]

Responsible for the design, development, and maintenance of all the group's software infrastructure, including the deployment and automation of key Linux-based services.

Applications ranged from internal financial administration and reporting tools, project management tools, internal infrastructure monitoring, and data exporting, to client-facing CMS building and customisation.

Education

University of Huddersfield, UK

- BSc Software Development & Multimedia (1st Class), 2008

Talks

A Dynamic, Statically Typed Contradiction

Given at ClojureX, London, Dec 2017 and :clojured, Berlin, Feb 2018.

Speculative Development

Given at ClojureX, London, Dec 2016 and :clojured, Berlin, Feb 2017.